

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

A wire with a length of **106** inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is **6** more than **4** times the value of y . What is the value of x ?

Answer

- A. **25**
- B. **28**
- C. **56**
- D. **86**

Correct Answer: D

Rationale

Choice D is correct. It's given that a wire with a length of **106** inches is cut into two parts. It's also given that one part has a length of x inches and the other part has a length of y inches. This can be represented by the equation $x + y = 106$. It's also given that the value of x is **6** more than **4** times the value of y . This can be represented by the equation $x = 4y + 6$. Substituting $4y + 6$ for x in the equation $x + y = 106$ yields $4y + 6 + y = 106$, or $5y + 6 = 106$. Subtracting **6** from each side of this equation yields $5y = 100$. Dividing each side of this equation by **5** yields $y = 20$. Substituting **20** for y in the equation $x = 4y + 6$ yields $x = 4(20) + 6$, or $x = 86$.

Choice A is incorrect. This value represents less than half of the total length of **106** inches; however, x represents the length of the longer part of the wire, since it's given that the value of x is **6** more than **4** times the value of y .

Choice B is incorrect. This value represents less than half of the total length of **106** inches; however, x represents the length of the longer part of the wire, since it's given that the value of x is **6** more than **4** times the value of y .

Choice C is incorrect. This represents a part that is **6** more than the length of the other part, rather than **6** more than **4** times the length of the other part.